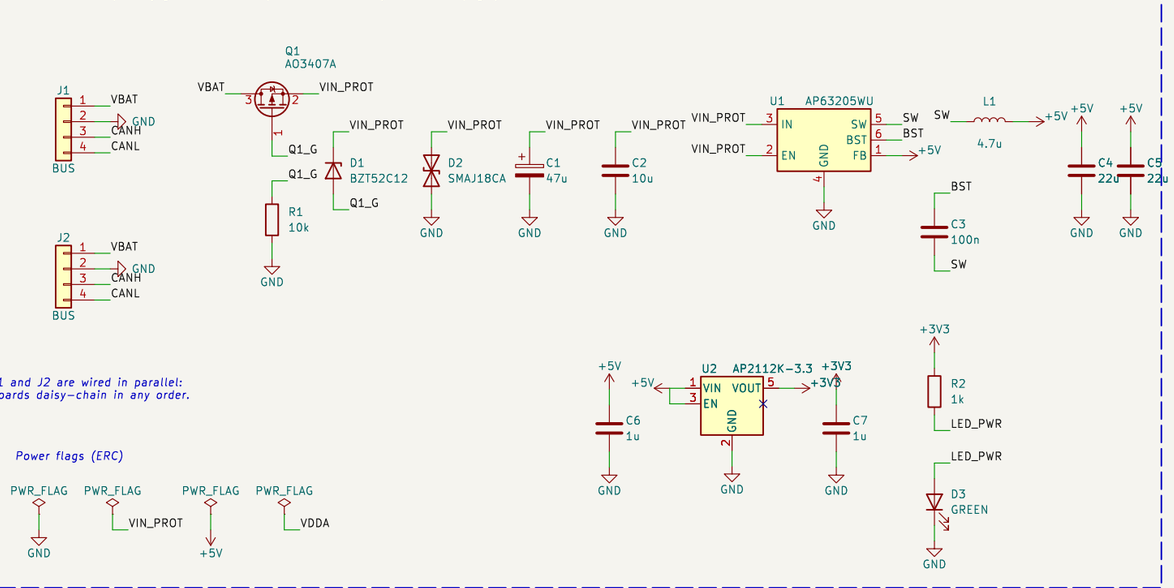
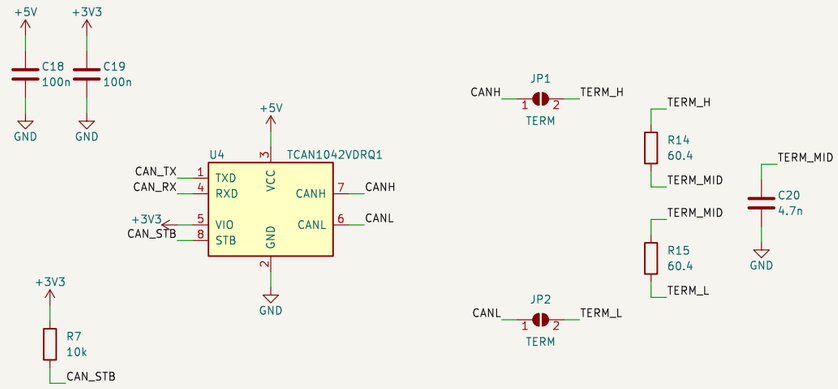
## POWER

*6–16 V in, reverse-polarity guard, 5 V buck (servo, sensor, CAN), 3.3 V LDO (logic)*



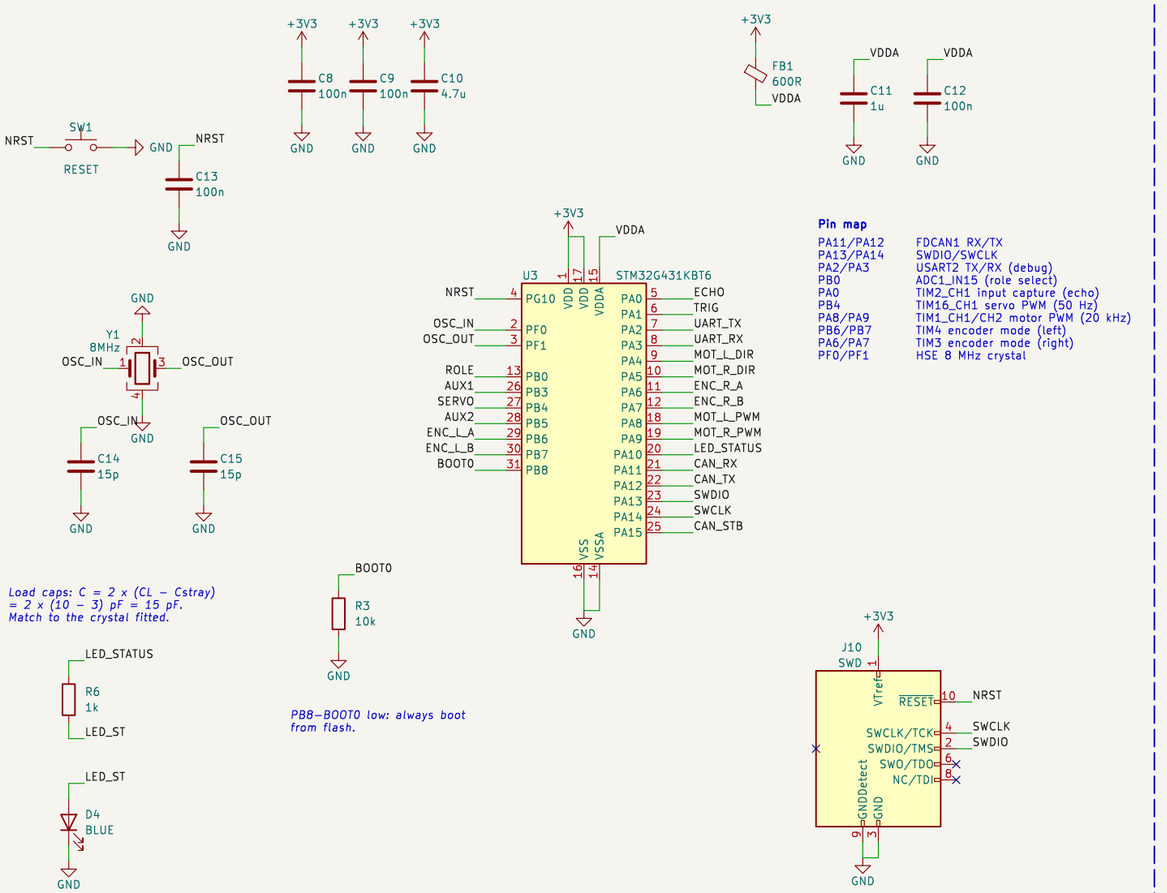

## CAN

*Transceiver on 5 V, logic side on 3.3 V (VIO); jumper-selected split termination*



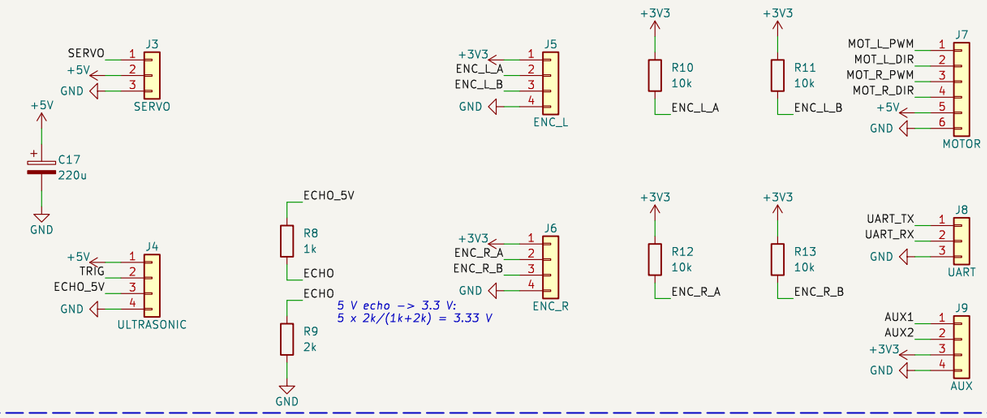

## MCU

*STM32G431KBT6, 8 MHz crystal for CAN timing, SWD programming*



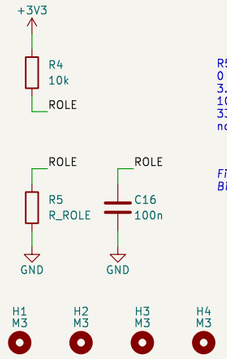

## I/O CONNECTORS

*All headers fitted on every board; the role decides which ones the firmware uses*




## ROLE SELECT

*One ADC pin + one resistor = one PCB and one firmware image for every node*




| R5 fitted | V(ROLE) | Role |
|---|---|---|
| 0 R | 0.00 V | Sensor node (ultrasonic + servo) |
| 3.3 k | 0.82 V | Receive node |
| 10 k | 1.65 V | Motor node (encoders + motor driver) |
| 33 k | 2.53 V | Spare |
| not fitted | 3.30 V | Safe mode: CAN listen-only, outputs off |

*Firmware averages 8 ADC samples at boot and picks the nearest bin.*
*Bins are ~0.8 V apart, so 5 % resistors never land in the wrong one.*

H1 M3 H2 M3 H3 M3 H4 M3

Input 6–16 V, CAN 500 kbit/s, STM32G431KBT6
One PCB for sensor, receive and motor nodes; role set by R5

**Parking robot CAN bus**

Sheet: /
File: can-node.kicad_sch

**Title: CAN Node Board**

| Size: A3 | Date: 2026-10-14 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/1 |